Project: Multilabel Image Classification

In *Chapter 2*, we extracted and stored our raw image data in our volume. In this chapter, we prepare our image dataset and save training and validation sets into Delta tables. To follow along in your workspace, please open the following notebook:

- Ch3-01-Loading Images 2 DeltaTables

We start by creating variables and removing existing data if the **Reset** widget value is "True".

```
delta_train_name = "train_imgs_main.delta"
delta_val_name = "valid_imgs_main.delta"

if bool(dbutils.widgets.get('Reset')):
  !rm -rf {main_dir_2write}{delta_train_name}
  !rm -rf {main_dir_2write}{delta_val_name}
```

Figure 3.29 – Cleaning up existing data if Reset == True

Next, we create a function to ingest all our images in one table. Initially, each label is in its own `folder_label_name`. We extract the `image_name`, `image_id`, and `label_id`, as well as create a `label_name` . Then we write the DataFrame as Delta.

```
from pyspark.sql import functions as f
def prep_data2delta(dir_name,outcomes,name2write,path2write="YOUR_PATH",write2delta=True,returnDF=None,):
    mapping_dict = {"buildings": 0,"sea": 1,"glacier": 2,"forest": 3,"street": 4,"mountain": 5,}
    # As we have multi label problem we will loop over labels to save them all under 1 main training set
    for label_name in outcomes:
        df = (
            spark.read.format("binaryfile")
            .option("recursiveFileLookup", "true")
            .load(f"{dir_name}/{lable_name}")
            .withColumn("label_name", f.lit(f"{label_name}"))
            .withColumn("label_id", f.lit(f"{mapping_dict[label_name]}").astype("int"))
            .withColumn("image_name", f.split(f.col("path"), "/").getItem(10))
            .withColumn("id", f.split(f.col("image_name"), ".jpg").getItem(0).astype("int"))
            )
        if write2delta:
            df.write.format("delta").mode("append").save(f"{path2write}{name2write}")
        if returnDF:
            return df
```

Figure 3.30 – The prep_data2delta function definition